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Cybersecurity Lab

Intelligent Network Traffic Monitoring System

DDoS Attack Detection & Real-Time Flow Classification Using Stacking Ensemble Learning

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SCAN GITHUB REPO

INTRODUCTION

- **DDoS Attacks Surge:** Malicious traffic floods overwhelm servers, causing critical service downtime.
- **Rule-Based Failure:** Traditional firewalls fail against dynamic, high-volume multi-vector DDoS.
- **Machine Learning Power:** Supervised ML learns subtle, multi-dimensional traffic flow behaviors automatically.

? RESEARCH HYPOTHESIS:
Can Stacking Ensemble minimize False Negatives while preserving 99.9%+ classification accuracy?

METHODS & EXPERIMENTAL WORKFLOW

STEP 1: Data Engineering
- 17 CIC-DDoS2019 Parquet files
- 426,076 Cleaned flow records
- Dropped nulls & duplicate rows
- Binary Labels: BENIGN (0), ATTACK (1)

STEP 2: Feature Optimization
- 88 -> 30 Features (RF Importance)
- Stratified 80/20 train/test split
- StandardScaler fitted on train set
- SMOTE balanced 530,194 samples

STEP 3: Ensemble & Inference
- 5 Models: DT, RF, XGB, Voting, Stacking
- Meta-Learner: Logistic Regression
- 99.94% accuracy on held-out test set
- Flask Dashboard for CSV batch demo

RESULTS & SYSTEM PERFORMANCE DEMONSTRATION

OBJECTIVE 1: Flow Feature Selection

Random Forest importance ranking reduced 88 -> 30 features.

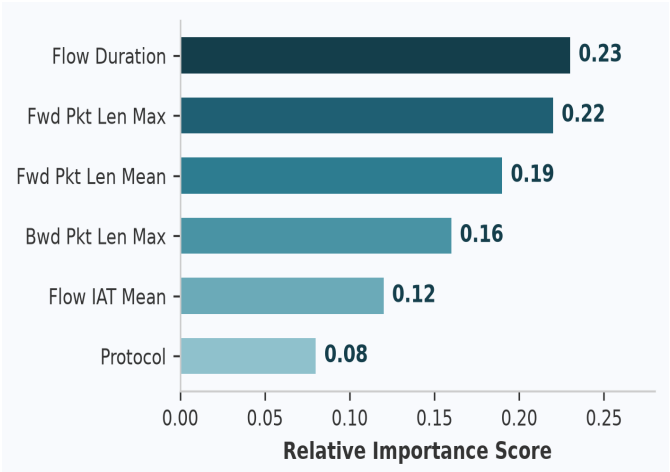


Figure 1. Relative Importance Score (Top 6 Selected Features)

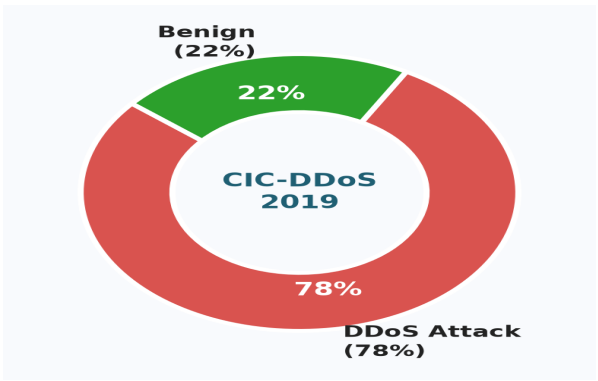


Figure 2. CIC-DDoS2019 Cleaned Traffic Balance (426K flows)

> Concluding Outcome 1:
Selecting top 30 features achieved 66% dimension reduction while retaining 100% predictive power and boosting speed.

OBJECTIVE 2: Detection Architecture

Multi-tier ensemble processing & real-time classification.

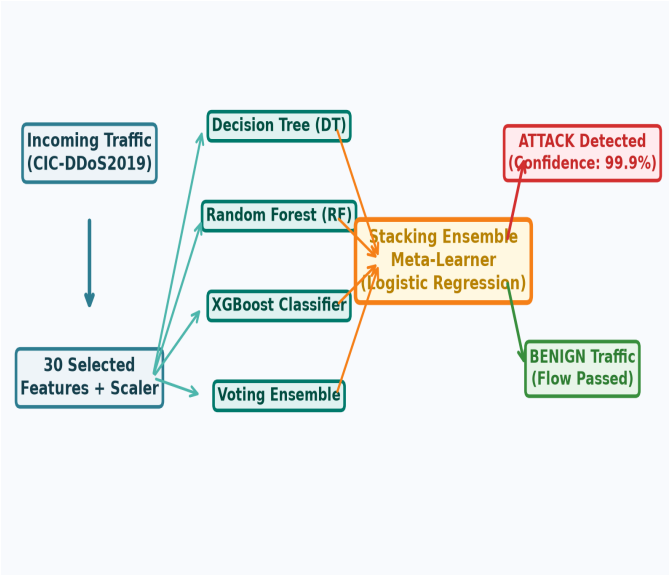


Figure 3. Intelligent Traffic Monitoring & Stacking Workflow

1. **Base Learners Layer:** DT, RF, and XGBoost evaluate incoming flow vectors simultaneously.
2. **Meta-Learner Engine:** Logistic Regression weighs individual model outputs to minimize errors.
3. **Web Dashboard Inference:** Flask application serves instant classification with confidence score.

> Concluding Outcome 2:
Stacking meta-learner dynamically resolves edge cases, eliminating single-model bias and providing robust inference.

OBJECTIVE 3: Benchmark & Evaluation

Testing across 85,216 unseen held-out network traffic flows.

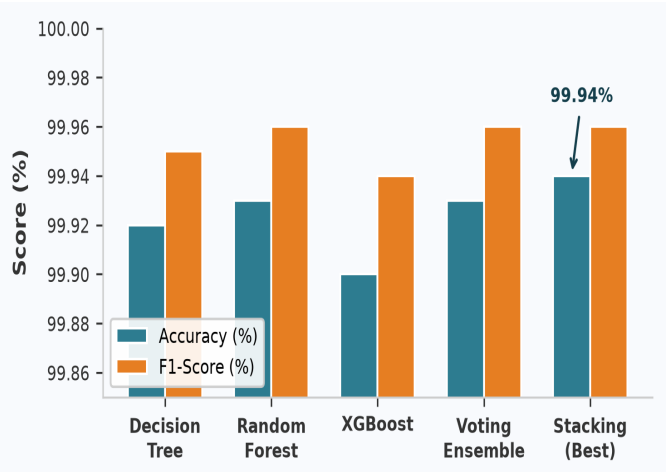


Figure 4. Accuracy & F1-Score Comparison Across 5 Models

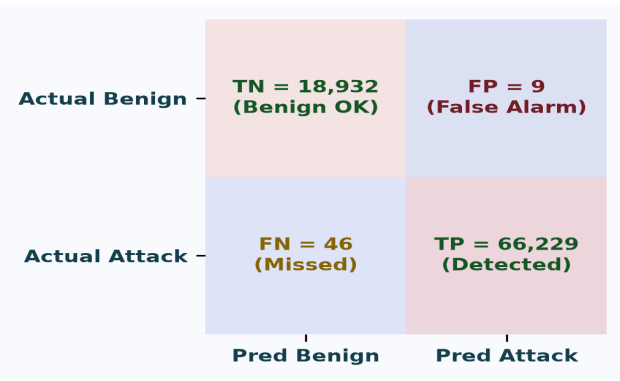


Figure 5. Stacking Confusion Matrix (Test Size = 85,216 flows)

> Concluding Outcome 3:
Stacking Ensemble achieved lowest False Negatives (FN = 46) and only 9 False Positives out of 85,216 test traffic flows.

DISCUSSION & KEY STUDY FINDINGS

99.94% Accuracy
Highest score across 85,216 test flows

FN = 46 (Lowest)
Missed rate < 0.07%, high security

FP = 9 (Minimal)
Only 9 false alarms in 18,941 normal

Cybersecurity Interpretation:
* Minimizing False Negatives is critical: missing an active DDoS flood allows server crashes.
* High precision (FP=9) prevents blocking legitimate university or enterprise web users.

Study Limitations:
* Offline flow processing only (no kernel packet capture / Scapy hook implemented).
* Binary classification only (identifies attack vs benign, not specific DDoS sub-type).

CONCLUSION & FUTURE SCOPE

Summary of Findings:
> Stacking Ensemble outperformed all individual base classifiers.
> SMOTE effectively resolved 78:22 class imbalance without test leakage.

Future Directions:
> Integration with Scapy/eBPF for live network packet interception.
> Multi-class extension to pinpoint 11 attack sub-types (SYN, UDP, etc.).

TAKE-HOME MESSAGE:
Ensemble meta-learning delivers optimal sensitivity and reliability, making AI-driven network protection feasible.

References & Open Source Repository:

[1] Sharafaldin et al., 'Developing Realistic DDoS Datasets' (CIC-DDoS2019).
[2] Pedregosa et al., 'Scikit-learn: Machine Learning in Python' (JMLR).
[3] GitHub: github.com/JainAditi123/Intelligent-Network-Traffic-Monitoring-System